

# Harsath Kaliappa Gounder Thangavel

☎ 647-619-3864 ✉ harsath@fastmail.com 🐙 github.com/harsath 🌐 harsath.net 🔗 linkedin.com/in/harsath

## Education

### York University

Bachelor of Science in Computer Science (Honours)

Toronto, Canada

January 2020 – December 2024

## Work Experience

### Privex

Software Engineer

October 2023 – January 2024

London, United Kingdom

- Developed an AI customer service agent using RAG techniques on a self-hosted LLM to handle common customer inquiries related to technical issues, products, and services.
- Developed REST APIs in Flask for querying and training the LLM, integrated with Discord APIs to allow customers to query the AI agent and staff to train it with new data on demand.
- Utilized vector databases and MySQL for training data management. Developed scripts to remove existing data from the database and to add new data from the website's documentation.
- Developed systemd service files and scripts to monitor and automatically reboot for error recovery.

### CogniPet

Software Engineer

April 2019 – December 2019

Zürich, Switzerland

- Developed Pet Breed Detection system by training a deep learning model using the Xception neural network and transfer learning techniques in Keras to recognize 120 breeds of dogs.
- Implemented REST APIs in Flask for model inference and integrated them with CogniPet's Android and iOS apps using Amazon Web Services (AWS).
- Experimented with video-based techniques for a pet facial recognition system by collecting data from dogs and analyzing features from multiple frames.

## Projects

### Jix Interpreter | C Interpreter

- Developed an interpreter for a dynamically typed programming language.
- Implemented fault-tolerant parsing and error handling, arrays, function pointers, loops, nested blocks, recursion, built-in functions, conditional statements, and control flow statements.
- Implemented basic functional programming support, including first-class and higher-order functions.
- Tested using automated tests with CI/CD pipelines.

### Prism Coverage | Java Code Coverage Engine

- Developed a code coverage engine using ANTLR4 for a programming language.
- Implemented control flow coverage (statement, decision, condition, path) criteria.
- Implemented data-flow coverage (All-Defs, All-DU-Paths, All-P-Uses) criteria.
- Implemented color-coded HTML report generation for the coverage engine.

### Cap Containers | C Containers Library

- Developed a highly generic, header-only containers library.
- Implemented 12 data structures, containers, and caches with iterator support.
- Implemented thread-safe variants with cross-platform support.
- Tested using automated tests with CI/CD pipelines.

### SOCKS5 Handler | C++ SOCKS5 Library

- Developed a SOCKS5 proxy library with dual-stack IPv4 and IPv6 support.
- Implemented RFC 1928 and RFC 1929 standards.

## Technical Skills

**Languages:** Python, C++, C, Java, SQL, JavaScript, Go, Perl, Bash, PHP, HTML/CSS,  $\text{\LaTeX}$

**Technologies:** Linux/Unix, MySQL, PostgreSQL, Docker, Microservices, Redis, MongoDB, Django, Flask, JWT, CI/CD, Neo4j

**Developer Tools:** Git, LXC/LXD, GDB, Amazon Web Services